Math 477 Homework 6

Spring 2026
Rutgers University

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Question 1:

Let S_{1000} denote the stock's price after 1000 periods and $X \sim B(1000, 0.52)$ denote the number of upwards price movements.

Then,

$$S_{1000} = s \cdot u^X \cdot d^{1000-X}$$

Substituting for our initial conditions, we have

$$S_{1000} = s \cdot 1.012^X \cdot 0.990^{1000-X} \geq 1.30s$$

Dividing through by s, we get

$$\begin{aligned} 1.012^X \cdot 0.990^{1000-X} &\geq 1.30 \\ X \ln|1.012| + (1000 - X) \ln|0.990| &\geq \ln|1.30| \\ X(\ln|1.012| - \ln|0.990|) + 1000 \ln|0.990| &\geq \ln|1.30| \\ X &\geq \frac{\ln|1.30| - 1000 \ln|0.990|}{\ln|1.012| - \ln|0.990|} \end{aligned}$$

Approximating, we get

$$X \geq 469.21.$$

Thus, we need $X \geq 470$ in order for $S_{1000} \geq 1.3s$.

$$\begin{aligned} P(X \geq 470) &= P\left(\frac{X - 1000(0.52)}{\sqrt{1000(0.52)(0.48)}} \geq \frac{470 - 1000(0.52)}{\sqrt{1000(0.52)(0.48)}}\right) \\ &= 1 - \Phi(-3.20) \\ &\approx 0.999 \end{aligned}$$

Question 2:

(a) We need $\int_0^\infty \int_{-y}^{y} c(y^2 - x^2)e^{-y}\ dxdy = 1$. Solving the integral:

$$\begin{aligned} c\int_0^\infty e^{-y}\int_{-y}^{y} y^2 - x^2\ dxdy &= c\int_0^\infty e^{-y}\left[y^2x - \frac{x^3}{3}\right]_{-y}^{y}\ dy \\ &= \frac{4c}{3}\int_0^\infty y^3e^{-y}\ dy \\ &= \frac{4c}{3}\cdot\Gamma(4) \\ &= 8c \\ &= 1 \end{aligned}$$

Thus, we have that $c = \frac{1}{8}$.

(b) For y:

$$\begin{aligned} f_Y(y) &= \int_{-y}^{y} f(x,y)\ dx \\ &= \frac{1}{8}e^{-y}\int_{-y}^{y}(y^2 - x^2)\ dx \\ &= \frac{y^3e^{-y}}{6} \end{aligned}$$

For x:

$$\begin{aligned} f_X(x) &= \int_{|x|}^{\infty} f(x,y)\ dy \\ &= \frac{1}{8}\int_{|x|}^{\infty}(y^2 - x^2)e^{-y}\ dy \\ &= \frac{1}{8}\int_{|x|}^{\infty} y^2e^{-y} - \frac{x^2}{8}\int_{|x|}^{\infty} e^{-y}\ dy \\ &= \frac{1}{8}e^{-|x|}(x^2 + 2|x| + 2) - \frac{x^2}{8}e^{-|x|} \\ &= \frac{e^{-|x|}(1 + |x|)}{4} \end{aligned}$$

(c)

$$\begin{aligned} E[X] &= \int_{-\infty}^{\infty} xf_X(x)\ dx \\ &= \int_{-\infty}^{\infty} x\cdot\frac{e^{-|x|}(1 + |x|)}{4} \end{aligned}$$

Notice that the integrand is a odd function, say, $g(x)$:

$$g(-x) = -x\cdot\frac{e^{-|x|}(1 + |x|)}{4} = -g(x).$$

Since we are integrating over a symmetric interval, both sides cancel out. Thus,

$$E[X] = 0$$

Question 3:

Let $X \sim U(15, 45)$ denote the man's arrival time in minutes after 12, and $Y \sim U(0, 60)$ denote the same for the woman.

(a)

$$\begin{aligned}
P(|X - Y| \leq 5) &= 1 - P(|X - Y| > 5) \\
&= 1 - (P(X - Y > 5) + P(Y - X > 5)) \\
&= 1 - \frac{1}{1800} \int_{15}^{45} (x - 5) + (60 - (x + 5)) dx \\
&= 1 - \frac{1}{1800} \int_{15}^{45} 50 dx \\
&= 1 - \frac{1500}{1800} \\
&= \frac{1}{6}
\end{aligned}$$

(b)

$$\begin{aligned}
P(X < Y) &= \frac{1}{1800} \int_{15}^{45} (60 - x)\ dx \\
&= \frac{1}{1800} \left[60x - \frac{x^2}{2} \right]_{15}^{45} \\
&= \frac{1}{2}
\end{aligned}$$